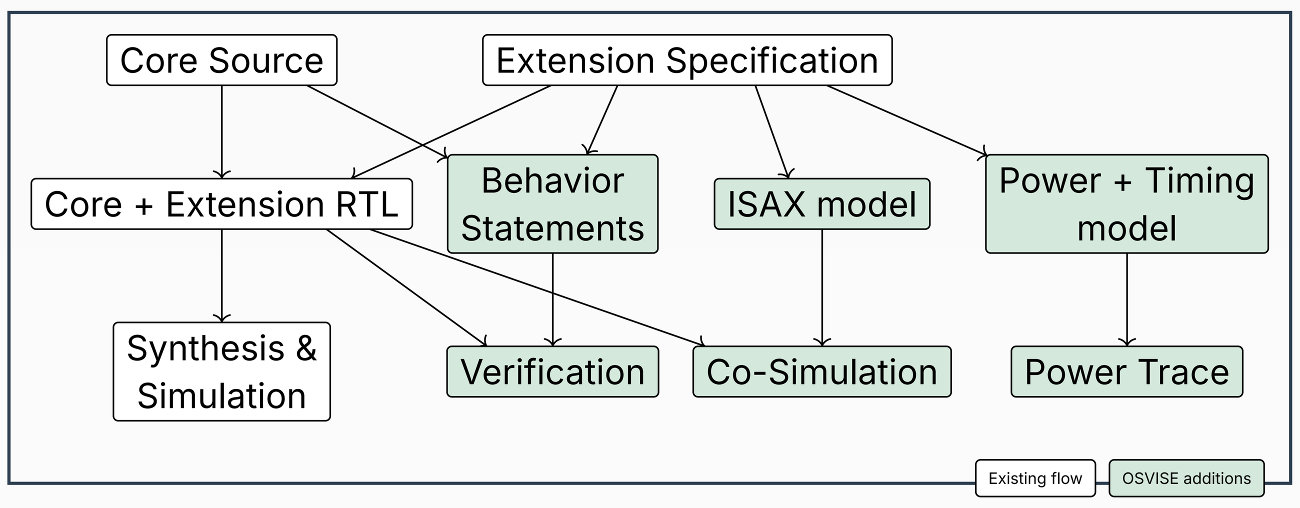
DI-OSVISE

Open Source Verification of Instruction Set Extensions

Overview

OSVISE is motivated by the need for open-source tools for verification of instruction set extensions. RISC-V enables frictionless instruction set extensions. This requires an ecosystem which supports fast evaluation, verification, and investigation of nonfunctional properties.

The OSVISE project focuses on closing the gaps in the open-source EDA ecosystem when working on Instruction Set Extensions. Central to this tool stacks are: **CIRCT**, **Yosys**, **Verilator**, **ETISS**, **CoreDSL**.



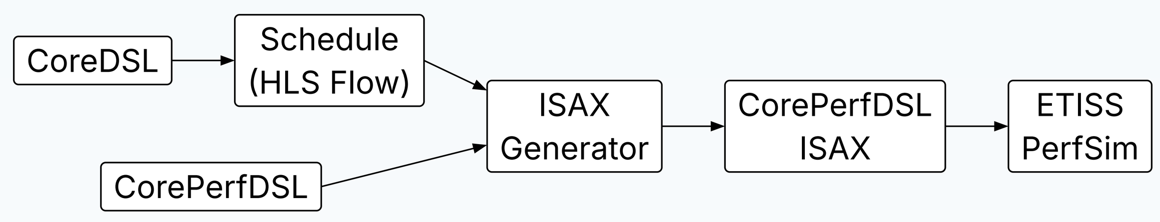
Simulation Framework

- Extending Verilator UVM capabilities
- UVM testbench for instruction extensions with tracing

Status:
- Verilator: randomization and constraint handling
- Equivalent verification results for example extension between commercial tools and Verilator
- Successful RISC-V CPU firmware simulation using extended Verilator

Performance Modeling

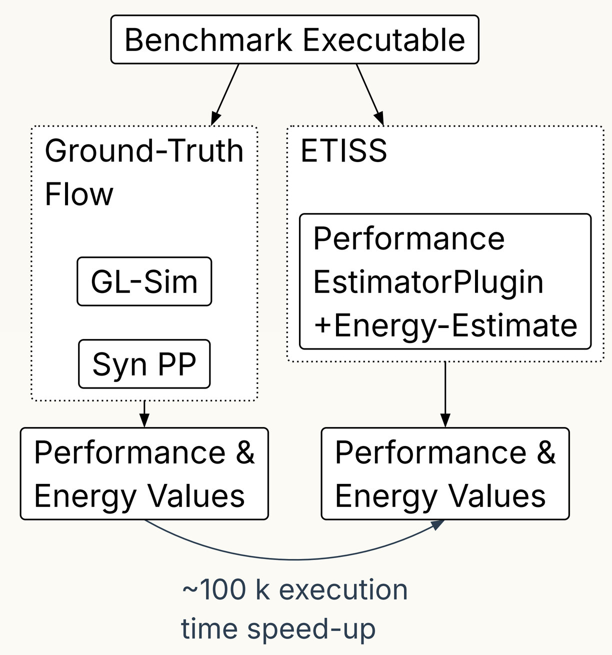
CorePerfDSL Generator:
- Reuses ISAX generator flow output
- Integrates into ETISS simulators for precise performance estimates



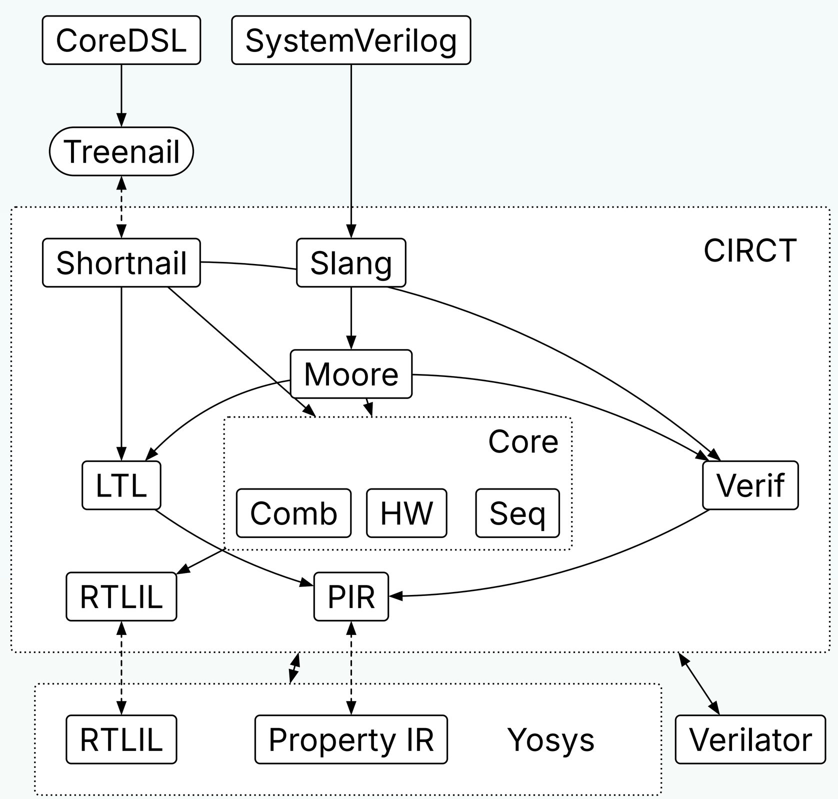
Status:
- Specification completed
- Prototype implemented

Nonfunctional Verification

Extendable Translating Instruction Set Simulator (ETISS) with Performance Estimation:
- Demonstrates high-level simulation feasibility using Co-Simulation
- Enables rapid performance and energy evaluation
- Estimates performance and energy of generated instruction streams using the core's average power



Power Estimation:
- Power estimation based on switching activity extracted from VCDs
- Waveform parsing using the open-source Waveform Analysis Language (WAL)
- Uses Liberty files for hardware parameters and netlists for gate-level analysis



Status:
- Demonstration for RISC-V dotprod ISAX
- CorePerfDSL Generator for ISAX
- Power estimation and modeling methods survey
- Open-source analytical power estimation model for RTL and gate-level

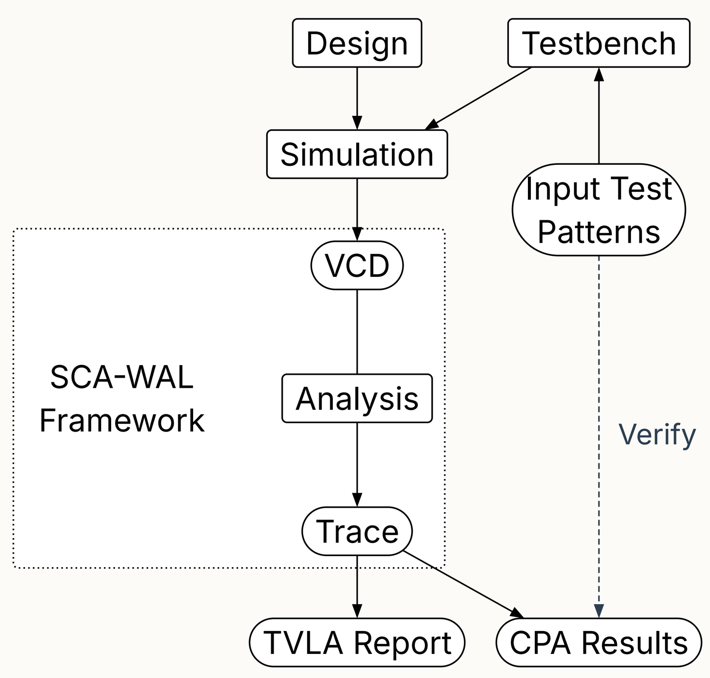
Instruction Stream

Instruction Stream Generator:
- Architectural-aware instruction stream for checks across multiple instructions
- Specification of constraints to describe architecture
- ISAX specified in CoreDSL

Status:
- Specification completed
- Prototype implemented

Functional Verification

SystemVerilog Assertions:
- Hardware verification is crucial, yet open-source tooling remains limited
- OSVISE extends open-source support for concurrent assertions
- CIRCT acts as a central hub, mapping diverse inputs to central *core* dialects via specific conversions



Instruction Extension
- Extract verification statements from instruction behavior
  - Create checker circuits from verification statements with transformations in Yosys

New Design Flows: Using CIRCT as an interchange framework enables novel methodologies. For example, *Chosys* provides a *Chisel*-to-*Yosys* path without requiring SystemVerilog.

Status:
- Prototype for CoreDSL to SVA conversion
- CIRCT to Yosys via RTLIL demonstrated
- Yosys Property IR defined

Open-Source Impact

- Development is done in the open
- Enables outside collaboration
- Allows for direct feedback

- Public access to new tools
  - tum-ei-eda/isaac-perf-gen
  - Minres/CRIG
  - planvtech/PlanV_Verilator_Feature_Tests
  - hm-aemy/coredsl-to-sva
  - hm-aemy/Shortnail
    - Extends: esa-tu-darmstadt/Shortnail
  - hm-aemy/chosys
  - YosysHQ/property-ir
  - tum-ei-eda/PerformanceSimulation_workspace
  - iti-luebeck/sca-wal/
    - Extends: ics-jku/wal
- Contributions to existing tools
  - llvm/circt
  - tum-ei-eda/etiss
  - YosysHQ/yosys
  - verilator/verilator

Evolving the open-source EDA ecosystem through ongoing contributions, ensuring immediate impact and continuous improvement.

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